VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



COMPUTER NETWORK REPORT

Homework 3b

Advisor(s):

Student(s):

HO CHI MINH CITY, NOVEMBER 2026

1 Question 1:

Question: Select the first UDP segment in your trace. What is the packet number4 of this segment in the trace file? What type of application-layer payload or protocol message is being carried in this UDP segment? Look at the details of this packet in Wireshark. How many fields there are in the UDP header?

Answer: The first UDP segment in the trace is packet number 5. It carries a SSDP (Simple Service Discovery Protocol) notification message, as indicated by the Info column (NOTIFY * HTTP/1.1). The UDP header contains four fields: Source Port, Destination Port, Length, and Checksum.

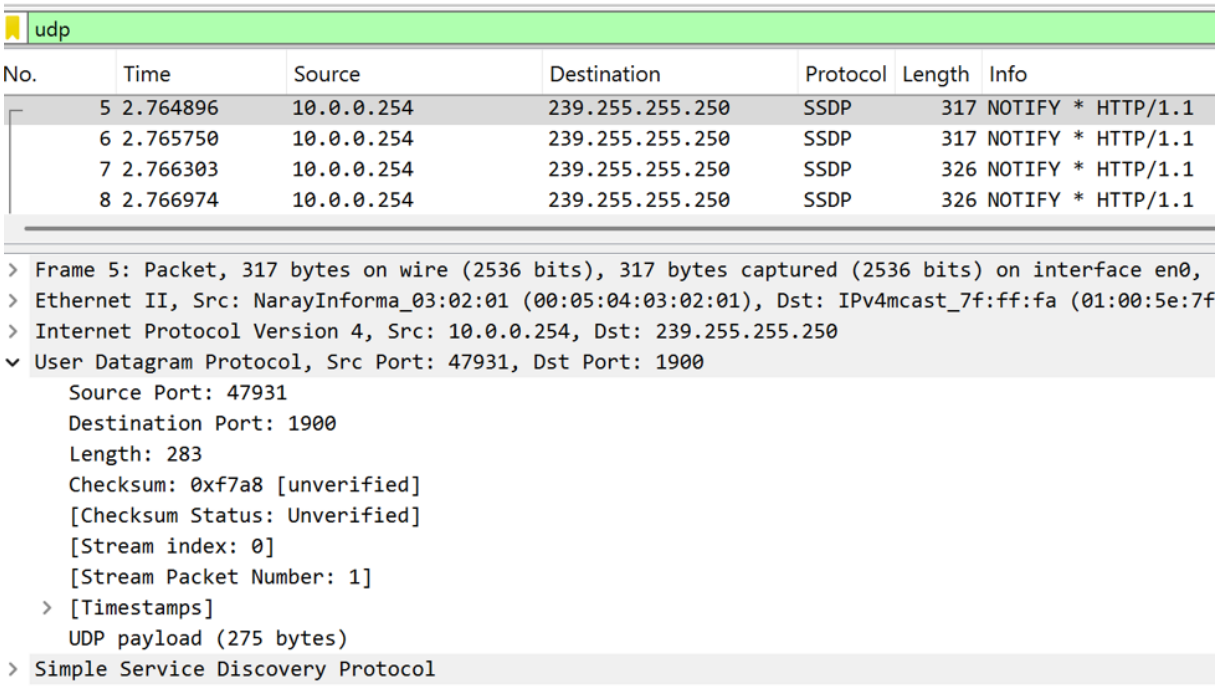


Figure 1.1

2 Question 2:

Question: By consulting the displayed information in Wireshark's packet content field for this packet (or by consulting the textbook), what is the length (in bytes) of each of the UDP header fields?

Answer: The UDP header contains four fields: Source Port, Destination Port, Length, and Checksum. Each field is 2 bytes (16 bits) long, making the total UDP header length 8 bytes (64 bits).



3 Question 3:

Question: The value in the Length field is the length of what?. Verify your claim with your captured UDP packet.

Answer: Looking at packet 5

- **UDP Length field:** 283 bytes
- **UDP header:** 8 bytes (4 fields $\times$ 2 bytes)
- **UDP payload:** 275 bytes

Verification: $8 + 275 = 283$
This matches the UDP Length field.

4 Question 4:

Question: What is the maximum number of bytes that can be included in a UDP payload?

Answer: The Length field is 2 bytes (16 bits), so its largest value is:

$$2^{16} - 1 = 65,535 \text{ bytes}$$

That value covers the UDP header plus the payload, so:

$$\text{Maximum UDP payload} = 65,535 - 8 \text{ (header)} = 65,527 \text{ bytes}$$

5 Question 5:

Question: What is the largest possible source port number?

Answer: The Source Port field is 2 bytes (16 bits), and each bit can either be 1 or 1, so so 16 bits can represent $2^16 = 65,536$ different values. So, the largest value is 65,535

6 Question 6:

Question: What is the protocol number for UDP? Give your answer in decimal notation.

Answer: The protocol number for UDP is 17

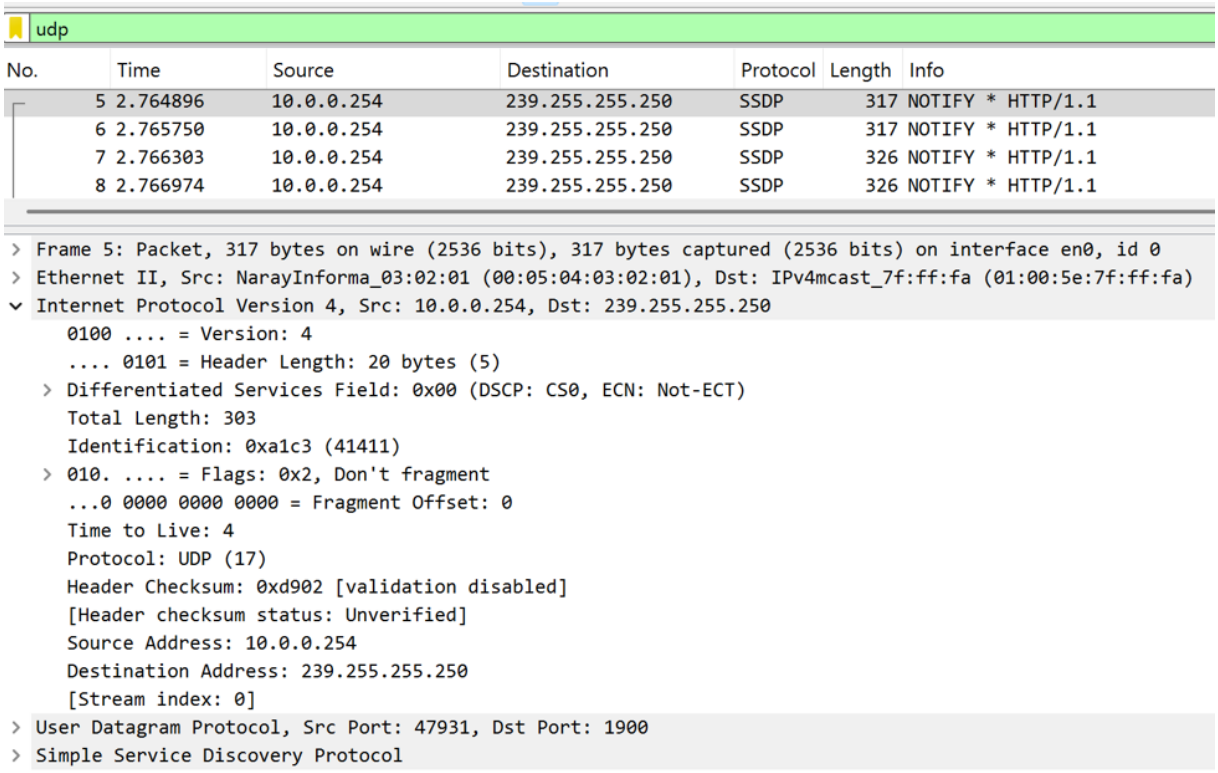


Figure 6.1

7 Question 7:

Question: Examine the pair of UDP packets in which your host sends the first UDP packet and the second UDP packet is a reply to this first UDP packet. What is the packet number5 of the first of these two UDP segments in the trace file? What is the value in the source port field in this UDP segment? What is the value in the destination port field in this UDP segment? What is the packet number6 of the second of these two UDP segments in the trace file? What is the value in the source port field in this second UDP segment? What is the value in the destination port field in this second UDP segment? Describe the relationship between the port numbers in the two packets.

Answer: The pair of packets is packet 15 (query), and packet 17 (response)

Packet 15 has source port at 58350, and destination port at 53
Packet 17 has source port at 53, and destination port at 58350
Relationship between both numbers: The source port of the first segment (58350) is the destination port of the second segment, and the destination port of the first segment (53) is the source port of the second segment. The ports are swapped. The reply goes back to the port the query came from, and the DNS server answers from port 53, the well-known DNS port it received the query on.

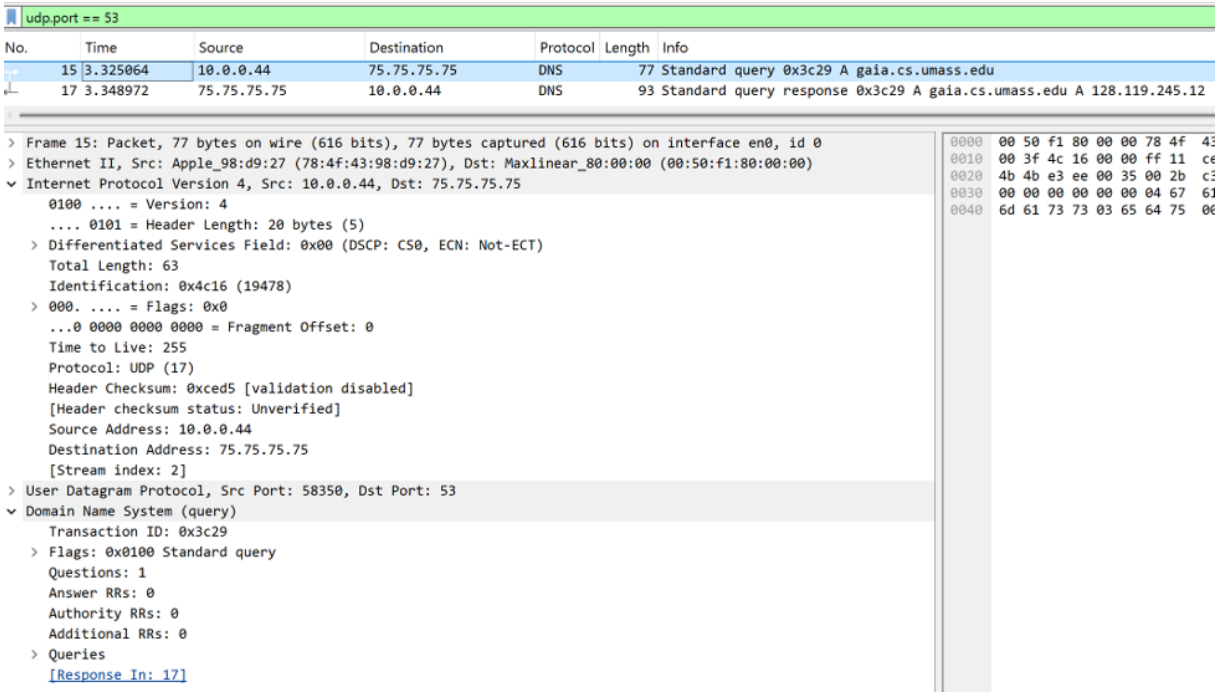


Figure 7.1

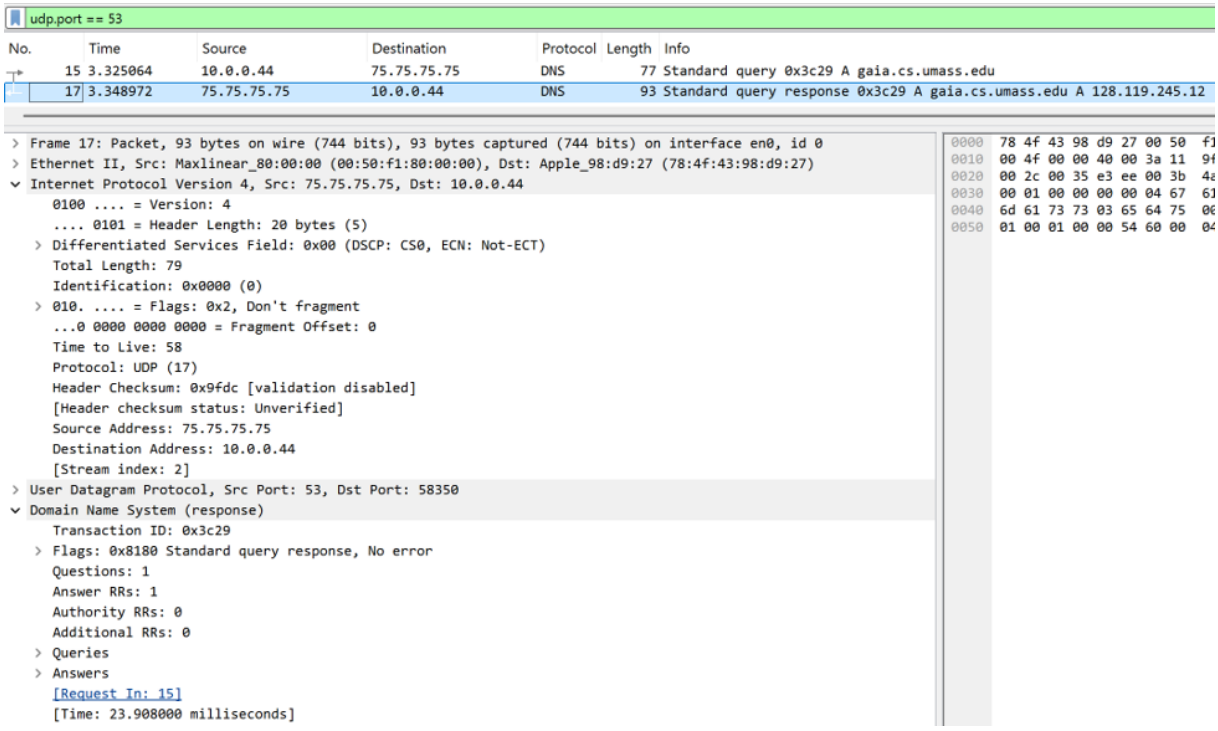

udp.port == 53
No. Time Source Destination Protocol Length Info
15 3.325064 10.0.0.44 75.75.75.75 DNS 77 Standard query 0x3c29 A gaia.cs.umass.edu
17 3.348972 75.75.75.75 10.0.0.44 DNS 93 Standard query response 0x3c29 A gaia.cs.umass.edu A 128.119.245.12
Frame 17: Packet, 93 bytes on wire (744 bits), 93 bytes captured (744 bits) on interface en0, id 0
Ethernet II, Src: Maxlinear_80:00:00 (00:50:f1:80:00:00), Dst: Apple_98:d9:27 (78:4f:43:98:d9:27)
Internet Protocol Version 4, Src: 75.75.75.75, Dst: 10.0.0.44
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 79
Identification: 0x0000 (0)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 58
Protocol: UDP (17)
Header Checksum: 0x9fdc [validation disabled]
[Header checksum status: Unverified]
Source Address: 75.75.75.75
Destination Address: 10.0.0.44
[Stream index: 2]
User Datagram Protocol, Src Port: 53, Dst Port: 58350
Domain Name System (response)
Transaction ID: 0x3c29
Flags: 0x8180 Standard query response, No error
Questions: 1
Answer RRs: 1
Authority RRs: 0
Additional RRs: 0
Queries
Answers
[Request In: 15]
[Time: 23.908000 milliseconds]
0000 78 4f 43 98 d9 27 00 50
0010 00 4f 00 00 40 00 3a 11
0020 00 2c 00 35 e3 ee 00 3b
0030 00 01 00 00 00 00 04 67
0040 6d 61 73 73 03 65 64 75
0050 01 00 01 00 00 54 60 00

Figure 7.2